Mechanical Design of a Simple Quadruped Robot

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Introduction

This report presents a preliminary mechanical design of a simple quadruped robot. The purpose of this design is to understand the basic concepts of robot mechanics, including body structure, leg design, motor selection, stability, and walking motion.

1. Body and Frame

The robot has a lightweight rectangular body made of aluminum or reinforced plastic with four legs. The battery and electronics are placed at the center of the body to improve balance.

2. Leg Design

Each leg consists of:

- Upper Leg
- Lower Leg

Each leg has two joints:

- Hip Joint
- Knee Joint

3. Joints and Degrees of Freedom (DOF)

Each leg has **2 DOF**, giving a total of:

$$\mathbf{4 \times 2 = 8\ DOF}$$

This is suitable for a simple walking robot.

4. Motor Selection

Servo motors are selected because they are easy to control, affordable, and provide sufficient torque for educational robot projects.

5. Preliminary Torque Calculation

Assume the robot weighs **4 kg**.

Each leg supports approximately **1 kg**.

Force:

$$\mathbf{F = m \times g = 1 \times 9.81 = 9.81\ N}$$

If the leg length is **0.1 m**:

$$\mathbf{T = F \times L = 9.81 \times 0.1 = 0.98\ N \cdot m}$$

Therefore, a motor with at least **1 N·m** torque is recommended.

6. Stability and Center of Gravity

The center of gravity should remain near the center of the body. Placing the battery in the middle improves stability during standing and walking.

7. Walking Method

The robot uses a diagonal walking gait by moving the front-right and rear-left legs together, followed by the front-left and rear-right legs.

8. Expected Mechanical Problems

- Limited motor torque.
- Leg slipping on smooth surfaces.
- Body vibration.
- High battery consumption.
- Joint wear over time.

Conclusion

This simple quadruped robot design meets the required mechanical design objectives and provides a good foundation for understanding basic robotics principles.